Integrated Research on Gravity and Fluid Dynamics: The H.U.G.G.E.R Framework, Adaptive Non-Linear Drag Coercivity, and SI Dimensional Homogeneity

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Abstract

This monograph presents the H.U.G.G.E.R (*Heuristic Universal Grid & Gravity Equilibrium Rendering*) continuum stress framework (v1.1–v1.2), advancing the macroscopic gravito-fluidic equilibrium model for rotating planetary atmospheres and oceans. While the v1.1 atmospheric module resolves polar coordinate singularities via unit-quaternion kinematics ($q_c \in \mathrm{Sp}(1)$) and suppresses supercritical turbulent kinetic energy via an adaptive non-linear drag tensor ($\boldsymbol{\mathcal{D}}_\kappa$), the v1.2 oceanic module establishes strict SI dimensional homogeneity ($[\mathrm{Pa} = \mathrm{kg} \cdot \mathrm{m}^{-1} \cdot \mathrm{s}^{-2}]$) and incompressible stress reduction ($\nabla \cdot \boldsymbol{u} = 0$). By coupling a thresholded power-law eddy viscosity $\kappa(|D\boldsymbol{u}|) = \kappa_0[1 + \alpha \max(0, |D\boldsymbol{u}| - s_c)^p]$ ($p \geq 1$) with the Virial tensor product divergence theorem, we prove a priori L^q ($q = p + 2 \geq 3$) strain-rate coercivity and finite-horizon enstrophy boundedness via Korn's inequality.

Contents

1 Introduction: Structural Bottlenecks in Macroscopic 3D Flow Simulations **2**

2 Governing Equation: Macroscopic Virial Stress Formulation **2**

3 CFD 1.1 Module: Adaptive Non-Linear Drag Suppression Tensor **2**

4 CFD 1.2 Module: Oceanic Incompressible Reduction and SI Homogeneity **3**
4.1 Incompressible Reduction and L^q ($q = p + 2 \geq 3$) Coercivity 3
4.2 Explicit Variable Definitions and SI Unit Homogeneity 3

5 Conclusion **3**

A Mathematical Bridge: Virial Tensor Product Divergence Theorem **3**

Acknowledgments and Declarations **4**

References **4**

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1 Introduction: Structural Bottlenecks in Macroscopic 3D Flow Simulations

In high-resolution numerical weather prediction (NWP) and macroscopic geophysical fluid dynamics (GFD) within a bounded planetary control volume $\Omega \subset \mathbb{R}^3$, classical continuum solvers confront two coupled instabilities [1, 3, 4]. First, spherical latitude–longitude grids suffer from metric convergence at the rotational poles ($\theta \in \{0, \pi\}$), inducing severe CFL time-step restrictions and coordinate singularities. Second, under intense Coriolis and baroclinic forcing, unresolved subgrid-scale vortex stretching triggers spectral energy pile-up and numerical matrix blow-up [2, 5].

To overcome these limitations, this study formulates the macroscopic Virial stress tensor $\mathbf{\Phi}_\Omega$ over a closed control boundary $\partial\Omega$ with volume normalization $V_\Omega = |\Omega| > 0$. By combining spatial mollification, unit-quaternion rotational stress kinematics, and an adaptive non-linear strain-rate drag tensor $\boldsymbol{\mathcal{D}}_\kappa$, the H.U.G.G.E.R framework bridges discrete grid-scale numerical stabilization and continuous macroscopic stress conservation without loss of physical or dimensional consistency.

2 Governing Equation: Macroscopic Virial Stress Formulation

The macroscopic thermodynamic stress functional governing momentum flux and Virial equilibrium across the boundary $\partial\Omega$ is defined by the volume-normalized surface integral:

$$\mathbf{\Phi}_\Omega = \frac{1}{V_\Omega} \mathrm{Re} \left\{ \oint_{\partial\Omega} \boldsymbol{r} \otimes \left(\mathbb{P}_{\mathrm{Kakeya}} : [\mathcal{K}_\omega \rho(r) \, (q_c \boldsymbol{\omega} q_c^*) \otimes (q_c \boldsymbol{\omega} q_c^*) + \Gamma_0 \boldsymbol{R}_\chi - P_{\lim} \boldsymbol{I} + \boldsymbol{T}_\chi] \right) \cdot \hat{\boldsymbol{n}} \, \mathrm{d}\Sigma \right\}. \quad (1)$$

Here, $\boldsymbol{r} \in \mathbb{R}^3$ is the spatial position vector, $\hat{\boldsymbol{n}}$ is the outward unit normal on $\partial\Omega$, $\mathcal{K}_\omega > 0$ ($[\mathrm{m}^2]$) is the angular momentum flux calibration factor, $\rho(r)$ ($[\mathrm{kg} \cdot \mathrm{m}^{-3}]$) is the stratified mass density, $q_c \in \mathrm{Sp}(1)$ is the unit rotation quaternion, and $\boldsymbol{\omega} = \nabla \times \boldsymbol{u}$ ($[\mathrm{s}^{-1}]$) is the vorticity of the velocity field $\boldsymbol{u}$. The rank-4 Kakeya directional projection tensor $\mathbb{P}_{\mathrm{Kakeya}}$ filters non-physical anisotropic spikes, while the hydrostatic limit pressure $-P_{\lim} \boldsymbol{I}$ and dense core stratification prevent compressive self-gravitational collapse, routing peripheral shear stresses along divergence-free slipstreams [8].

3 CFD 1.1 Module: Adaptive Non-Linear Drag Suppression Tensor

To actively regulate extreme turbulent kinetic energy (TKE) in compressible and weakly compressible planetary envelopes, the generic deviatoric efflux tensor $\boldsymbol{T}_\chi$ in Equation (1) is replaced by the explicit non-linear drag suppression tensor $-\boldsymbol{\mathcal{D}}_\kappa$:

$$\mathbf{\Phi}_{\mathrm{CFD\,1.1}} = \frac{1}{V_\Omega} \mathrm{Re} \left\{ \oint_{\partial\Omega} \boldsymbol{r} \otimes \left(\mathbb{P}_{\mathrm{Kakeya}} : \left[\mathcal{K}_\omega \rho(r) \, (q_c \boldsymbol{\omega} q_c^*)^{\otimes 2} + \Gamma_0 \boldsymbol{R}_\chi - P_{\lim} \boldsymbol{I} - \boldsymbol{\mathcal{D}}_\kappa \right] \right) \cdot \hat{\boldsymbol{n}} \, \mathrm{d}\Sigma \right\}. \quad (2)$$

Let $D\boldsymbol{u} := \frac{1}{2}(\nabla \boldsymbol{u} + (\nabla \boldsymbol{u})^\mathsf{T})$ denote the symmetric rate-of-strain tensor and $|\boldsymbol{\omega}| = |\nabla \times \boldsymbol{u}|$ the local vorticity magnitude. To preserve inviscid laminar accuracy below a critical strain/vorticity threshold $s_c > 0$ while guaranteeing superlinear ($p \geq 1$) dissipation in supercritical shear layers, the adaptive dynamic eddy viscosity κ is defined by:

$$\kappa(|D\boldsymbol{u}|) = \kappa_0 \left[1 + \alpha \max\big(0, |D\boldsymbol{u}| - s_c\big)^p \right], \quad (3)$$

where $\kappa_0 > 0$ ($[\mathrm{Pa} \cdot \mathrm{s}]$) is the baseline dynamic viscosity, $\alpha > 0$ ($[\mathrm{s}^p]$) is the non-linear amplification coefficient, and $p \geq 1$ is the Ladyzhenskaya–Smagorinsky power-law exponent [4,5]. The compressible deviatoric drag tensor is then given by:

$$\boldsymbol{\mathcal{D}}_\kappa = 2\kappa(|D\boldsymbol{u}|) \left(D\boldsymbol{u} - \frac{1}{3}(\nabla \cdot \boldsymbol{u}) \boldsymbol{I} \right) = \kappa(|D\boldsymbol{u}|) \left(\nabla \boldsymbol{u} + (\nabla \boldsymbol{u})^\mathsf{T} - \frac{2}{3}(\nabla \cdot \boldsymbol{u}) \boldsymbol{I} \right). \quad (4)$$

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4 CFD 1.2 Module: Oceanic Incompressible Reduction and SI Homogeneity

To validate the physical and mathematical consistency of the H.U.G.G.E.R stress closure, the v1.2 module specializes the governing equations to macroscopic incompressible oceanic circulation ($\nabla \cdot \boldsymbol{u} = 0$) [6,7].

4.1 Incompressible Reduction and L^q ($q = p + 2 \geq 3$) Coercivity

Under the incompressibility constraint $\nabla \cdot \boldsymbol{u} = \mathrm{tr}(D\boldsymbol{u}) = 0$, the volumetric trace term $-\frac{2}{3}(\nabla \cdot \boldsymbol{u})\boldsymbol{I}$ in Equation (4) vanishes identically, reducing $\boldsymbol{\mathcal{D}}_\kappa$ to the symmetric shear stress tensor:

$$\boldsymbol{\mathcal{D}}_\kappa^{\text{ocean}} = 2\kappa(|D\boldsymbol{u}|)D\boldsymbol{u} = \kappa(|D\boldsymbol{u}|)\left(\nabla \boldsymbol{u} + (\nabla \boldsymbol{u})^{\mathsf{T}}\right). \tag{5}$$

In the interior momentum balance $\partial_t \boldsymbol{u} + (\boldsymbol{u} \cdot \nabla)\boldsymbol{u} + \nabla P = \nabla \cdot \boldsymbol{\mathcal{D}}_\kappa^{\text{ocean}}$ on a periodic domain $\mathbb{T}^3$, testing with $\boldsymbol{u}$ yields the coercive energy dissipation rate $2\int_{\mathbb{T}^3} \kappa(|D\boldsymbol{u}|)|D\boldsymbol{u}|^2 \, \mathrm{d}x \geq c_1 \|D\boldsymbol{u}\|_{L^q}^q - c_2$ with $q = p + 2 \geq 3$. By Korn's inequality $\|\nabla \boldsymbol{u}\|_{L^q} \leq C_K \|D\boldsymbol{u}\|_{L^q}$, this $L^q(\mathbb{T}^3 \times (0,t))$ gradient bound controls the cubic vortex stretching integral $|\langle(\boldsymbol{\omega} \cdot \nabla)\boldsymbol{u}, \boldsymbol{\omega}\rangle| \leq \|\nabla \boldsymbol{u}\|_{L^3}^3$, preventing finite-time enstrophy blow-up:

$$\|\boldsymbol{\omega}(t)\|_{L^2(\mathbb{T}^3)}^2 \leq e^{2t}\left(\|\boldsymbol{\omega}_0\|_{L^2(\mathbb{T}^3)}^2 + 2C_q C_K^q \|D\boldsymbol{u}\|_{L^q(\mathbb{T}^3 \times (0,t))}^q\right) < \infty. \tag{6}$$

4.2 Explicit Variable Definitions and SI Unit Homogeneity

Table 1 establishes the exact correspondence between the tensor constituents of Equation (2) and their International System of Units (SI) mechanical dimensions.

Table 1: Physical correspondence and SI dimensional homogeneity of the H.U.G.G.E.R v1.2 stress constituents.

Tensor / Parameter	Geophysical / Continuum Correspondence	SI Unit / Dimension
$\boldsymbol{\sigma}_{\text{rot}} = \mathcal{K}_\omega \rho(r)(q_{\text{c}}\boldsymbol{\omega}q_{\text{c}}^*)^{\otimes 2}$	Unit-Quaternion Centrifugal / Rotational Stress	$[\text{Pa} = \text{kg} \cdot \text{m}^{-1} \cdot \text{s}^{-2}]$
$\boldsymbol{R}_{\text{X}}$	Baroclinic / Thermohaline Influx Stress Tensor	$[\text{Pa} = \text{kg} \cdot \text{m}^{-1} \cdot \text{s}^{-2}]$
$\boldsymbol{\mathcal{D}}_\kappa^{\text{ocean}}$	Adaptive Non-Linear Eddy Viscous Stress Tensor	$[\text{Pa} = \text{kg} \cdot \text{m}^{-1} \cdot \text{s}^{-2}]$
$P_{\text{lim}}\boldsymbol{I}$	Isotropic Hydrostatic Limit Pressure Tensor	$[\text{Pa} = \text{kg} \cdot \text{m}^{-1} \cdot \text{s}^{-2}]$
Γ_0	Thermodynamic Influx Linkage Coefficient	$[-]$ (Dimensionless)
$\mathcal{K}_\omega$	Angular Momentum Specific Area Calibrator	$[\text{m}^2]$

5 Conclusion

The H.U.G.G.E.R v1.1–v1.2 framework establishes a dimensionally homogeneous (Pa), singularity-free macroscopic stress closure for rotating continuum flows. By uniting unit-quaternion rotational tensors with L^q-coercive ($q = p + 2 \geq 3$) non-linear strain-rate dissipation, the model guarantees macroscopic enstrophy boundedness and provides the continuum foundation integrated into Part 1 of the 2026 *Grand Unified Framework* series.

A Mathematical Bridge: Virial Tensor Product Divergence Theorem

To rigorously connect the boundary-integrated Virial stress functional $\boldsymbol{\Phi}_\Omega$ in Equation (1) to the interior continuum momentum equations, we apply the Gauss divergence theorem componentwise.

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For any continuously differentiable total Cauchy stress tensor $\boldsymbol{\sigma} \in C^1(\bar{\Omega}; \mathbb{R}^{3\times 3})$ and position vector $\boldsymbol{r} = (x_1, x_2, x_3)^{\mathsf{T}}$:

$$\frac{1}{V_\Omega} \oint_{\partial\Omega} x_i \sigma_{jk} n_k \, \mathrm{d}\Sigma = \frac{1}{V_\Omega} \int_\Omega \partial_k (x_i \sigma_{jk}) \, \mathrm{d}V = \frac{1}{V_\Omega} \int_\Omega \left[\sigma_{ji} + x_i \partial_k \sigma_{jk}\right] \mathrm{d}V. \tag{7}$$

In invariant tensor notation, Equation (7) reads $\frac{1}{V_\Omega} \oint_{\partial\Omega} \boldsymbol{r} \otimes (\boldsymbol{\sigma} \cdot \hat{\boldsymbol{n}}) \, \mathrm{d}\Sigma = \frac{1}{V_\Omega} \int_\Omega \left[\boldsymbol{\sigma}^{\mathsf{T}} + \boldsymbol{r} \otimes (\nabla \cdot \boldsymbol{\sigma})\right] \mathrm{d}V$. This identity proves that macroscopic Virial boundary equilibrium necessitates interior stress divergence closure ($\nabla \cdot \boldsymbol{\sigma}$), directly linking macroscopic boundary observables to localized viscous and topological dissipation.

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Declarations

Data and Code Availability. All mechanized Lean 4 formal verification suites, symbolic CAS scripts, numerical solvers, and benchmark datasets supporting the findings of this study are publicly archived via the canonical repository (`https://github.com/arkhenomia/arkheion`) and the dataset archive (`https://huggingface.co/datasets/arkhenomia/arkheion`).

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